Soyabean Leaf Disease Detection Using Inception ResNet V2

A Deep Learning Approach for Automated Agricultural Disease Classification

Team Member

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Project Guide

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Introduction & Problem Statement

Why Soybean Matters

- One of the world's most critical oilseed crops — vital to food, feed, and industrial supply chains
- Leaf diseases can cause **20–40% yield loss** if undetected
- Early, accurate detection is essential for crop protection

The Core Problem

- Manual inspection by agronomists is **slow, costly,** and prone to human error
- Not scalable for large farmlands or remote regions
- Urgent need for an **automated, AI-driven** detection pipeline

Problem Statement

The Challenge

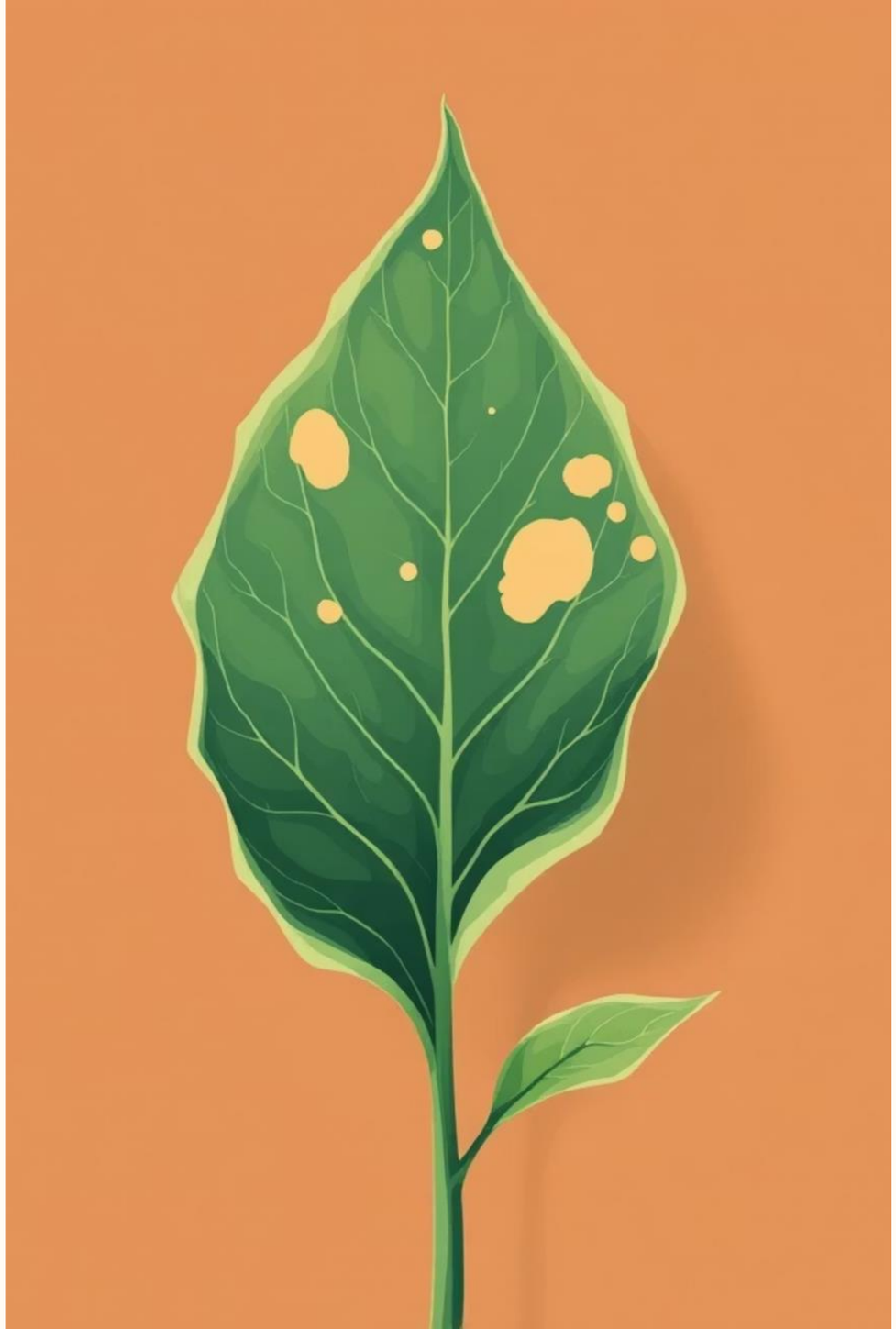
Soybean leaf diseases significantly reduce crop yield and quality.

Why Manual Detection Doesn't Work

- Time-consuming to inspect large fields
- Requires expert knowledge
- Not scalable for big farms

The Solution We Need

An automated, accurate, and scalable deep learning system to detect and classify soybean leaf diseases from images.



Objectives & Dataset

Project Objectives

Automated Detection

Detect soybean leaf diseases using deep learning with minimal human intervention

Multi-Class Classification

Accurately classify images across 8 distinct disease and healthy categories

Dataset Overview

Source

Zenodo Public Dataset

Classes

4 Disease Categories

Images

~10,000 Labelled Images

Split

70 / 15 / 15 Train/Val/Test

Model Selection & Improvement

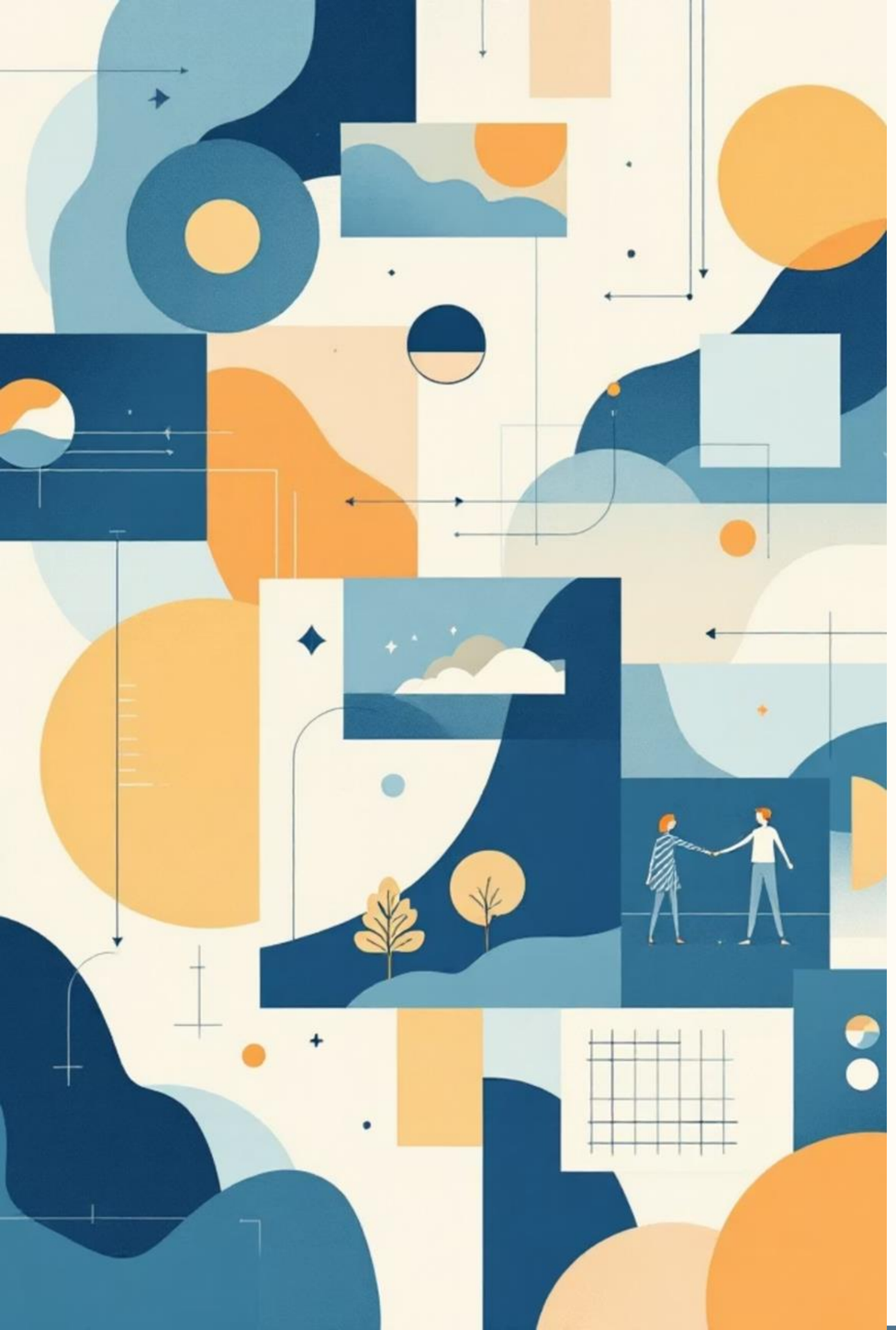
We Tested Different Models

MobileNetV2 (First Try)

- Lightweight and fast
- But: Overfitting and poor generalization on validation data

InceptionResNetV2 (Final Choice) ✓

- Combines deep feature extraction with multi-scale learning
- Better accuracy and generalization for complex plant disease patterns
- Best for capturing disease-specific details



Preprocessing & Augmentation

CLAHE — Contrast Enhancement

Contrast Limited Adaptive Histogram Equalisation improves local contrast in leaf images, making disease-affected regions more visually distinct for the model

Data Augmentation Techniques

Rotation · Horizontal/Vertical Flip · Zoom · Brightness Shift · Shear — applied on-the-fly during training to increase dataset diversity

Why It Matters

Augmentation prevents **overfitting**, simulates real-world imaging conditions, and improves the model's ability to generalise across unseen data

Model Architecture

InceptionResNetV2 — A Pre-Trained Deep Learning Model

Pre-Trained Foundation

The model was already trained on millions of images. We use this knowledge to recognize soybean leaves faster.

Feature Extraction

The model learns to identify important patterns in leaf images. It spots colors, shapes, and textures that indicate disease.

Attention Focus

A special attention module helps the model focus on diseased areas. It is like zooming in on the problem spots.

Fine-Tuning

We adjusted the model specifically for soybean diseases. This improves accuracy on our specific dataset.

Training Techniques

Adam Optimiser

Adaptive learning rate with momentum — ideal for sparse gradients in image data

LR Scheduling

ReduceLROnPlateau automatically decreases learning rate when validation loss stagnates

Early Stopping

Monitors validation loss and halts training to prevent overfitting and unnecessary computation

Class Weights

Compensates for **class imbalance** in the dataset — ensures minority disease classes are learnt effectively

Results

77%

Test Accuracy

Across all 8 disease classes on held-out test set

4

Classes Classified

Healthy + 3 distinct leaf disease categories(bacterial_blight ,cercospora_leaf_blight , frogeye, healthy)

10K

Training Images

Augmented and preprocessed dataset size

Performance Observations

- Model demonstrates **consistent classification** across most disease categories
- SE Attention module improved feature selectivity, contributing to accuracy gains over baseline InceptionResNetV2
- Class weighting reduced bias towards over-represented healthy class
- Accuracy of **77%** is a strong baseline for a PBL-scope deep learning project with limited compute

Application & Future Scope

Application (How It Works)

01

Upload a Leaf Image

02

Model Analyzes the Image

03

Get Instant Disease Prediction

Future Scope

Improve Accuracy

Use more training data and better techniques

Expand Dataset

Collect more leaf images from different regions

Mobile App

Make it easy to use on smartphones for farmers

Conclusion

> AI-powered vision systems can meaningfully contribute to food security by enabling **fast, scalable, and accurate** crop disease detection.

What We Built

An end-to-end deep learning system using InceptionResNetV2 + SE Attention for 4-class soybean disease classification

Key Outcome

Achieved 77% test accuracy with robust preprocessing, augmentation, and transfer learning on a ~10,000 image dataset

Thank You